

20260331 Proxy EW(H α)

1. Calculation of EW_{proxy}(H α)

Here we now compute EW_{proxy}(H α) from the observed H α line flux and the continuum-only cube `*_CONTcube.fits`, instead of using only the broadband *R*-band flux.

The observed H α flux map is taken from the `HA6562_FLUX` extension in either `*_gas_BIN_maps_extended.fits` or `*_gas_BIN_maps.fits`.

For the continuum window, we follow the [MaNGA DAP](#) H α EW passband in vacuum,

$$\lambda_{\text{vac,low}} = 6557.6 \text{ \AA}, \quad \lambda_{\text{vac,high}} = 6571.6 \text{ \AA}, \quad (1)$$

with the H α Ritz wavelength

$$\lambda_{\text{vac,ref}} = 6564.608 \text{ \AA}. \quad (2)$$

Because the local continuum cube wavelength axis is in standard air wavelength, these values are converted from vacuum to air using the NIST ASD / Peck & Reeder (1972) refractive-index formula,

$$\lambda_{\text{air}} = \lambda_{\text{vac}}/n, \quad (3)$$

with

$$(n - 1) \times 10^8 = 8060.51 + \frac{2480990}{132.274 - \sigma^2} + \frac{17455.7}{39.32957 - \sigma^2}, \quad (4)$$

where σ is the vacuum wavenumber in inverse microns.

The Ritz wavelength λ_{vac} in vacuum is equal to the inverse of σ . If σ is in units of cm^{-1} , and λ is in nanometers, then

$$\lambda_{\text{vac}} [\text{nm}] = \frac{10^7}{(\sigma [\text{cm}^{-1}])}. \quad (5)$$

This gives the rest-frame air H α reference wavelength and EW window used by the code.

The continuum cube is then shifted from observed frame to rest frame using the galaxy redshift z from `new_redshifts`:

$$\lambda_{\text{rest}} = \frac{\lambda_{\text{obs}}}{1 + z}. \quad (6)$$

At the same time, the observed continuum flux density is converted to rest-frame flux density per unit wavelength as

$$f_{\lambda, \text{rest}} = (1 + z) f_{\lambda, \text{obs}}. \quad (7)$$

The mean continuum flux density in the H α EW window is measured from all continuum-cube planes whose rest-frame wavelengths fall inside the selected air passband:

$$\bar{f}_{\lambda, \text{cont}} = \frac{1}{\Delta\lambda} \int_{\lambda_1}^{\lambda_2} f_{\lambda, \text{rest}}(\lambda) d\lambda. \quad (8)$$

In practice, this is evaluated as the sampled mean of the finite continuum-cube values inside the rest-frame H α window.

The new proxy equivalent width is therefore

$$\text{EW}_{\text{proxy}}(\text{H}\alpha) = \frac{F_{\text{H}\alpha}}{\bar{f}_{\lambda, \text{cont}}}, \quad (9)$$

For comparison, the old broadband-based pseudo EW is still preserved in a separate HDU. In that legacy approach, the observed R -band magnitude `MAGNITUDE_R_UNCORRECTED` from `*_SPATIAL_BINNING_maps_extended.fits` is first converted to nanomaggies,

$$R_{\text{nanomaggy}} = 10^{(22.5 - m_{R, \text{obs}})/2.5}, \quad (10)$$

then to flux density per unit frequency,

$$f_{\nu} = R_{\text{nanomaggy}} \times 3.631 \times 10^{-29} [\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1}], \quad (11)$$

and then to a single-wavelength continuum estimate at $6562.8 \text{ \AA}$,

$$f_{\lambda, \text{legacy}} = f_{\nu} c / \lambda^2. \quad (12)$$

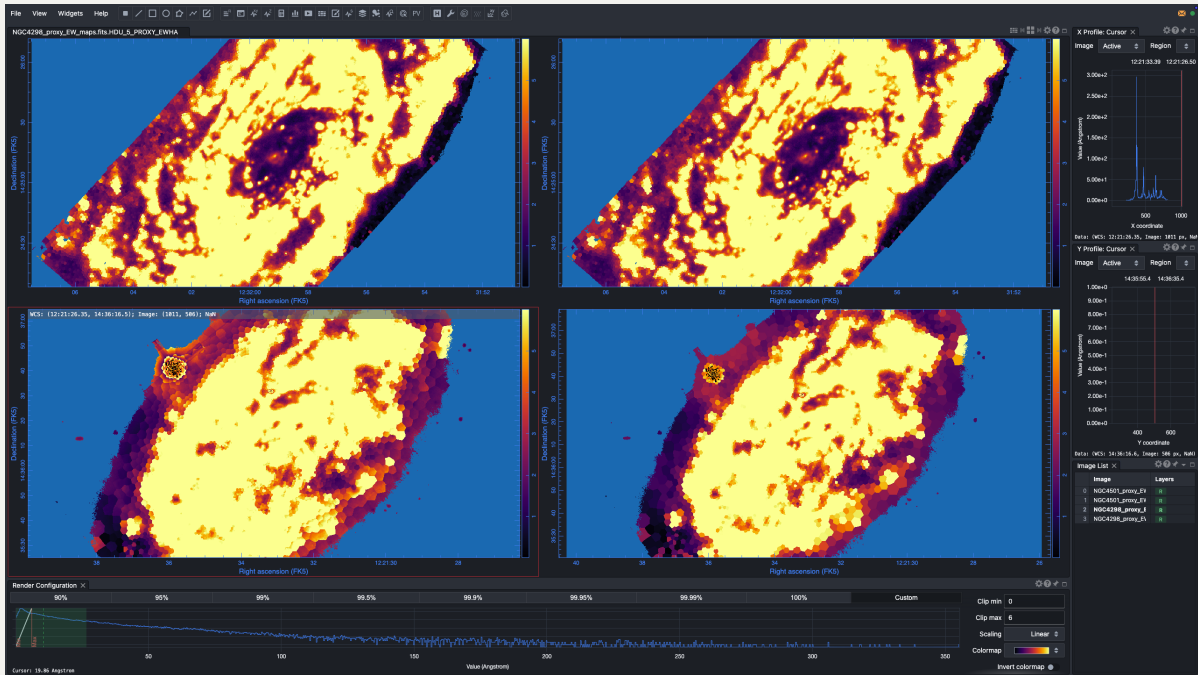
The old pseudo EW is therefore

$$\text{EW}_{\text{pseudo}}(\text{H}\alpha) = \frac{F_{\text{H}\alpha}}{f_{\lambda, \text{legacy}}}. \quad (13)$$

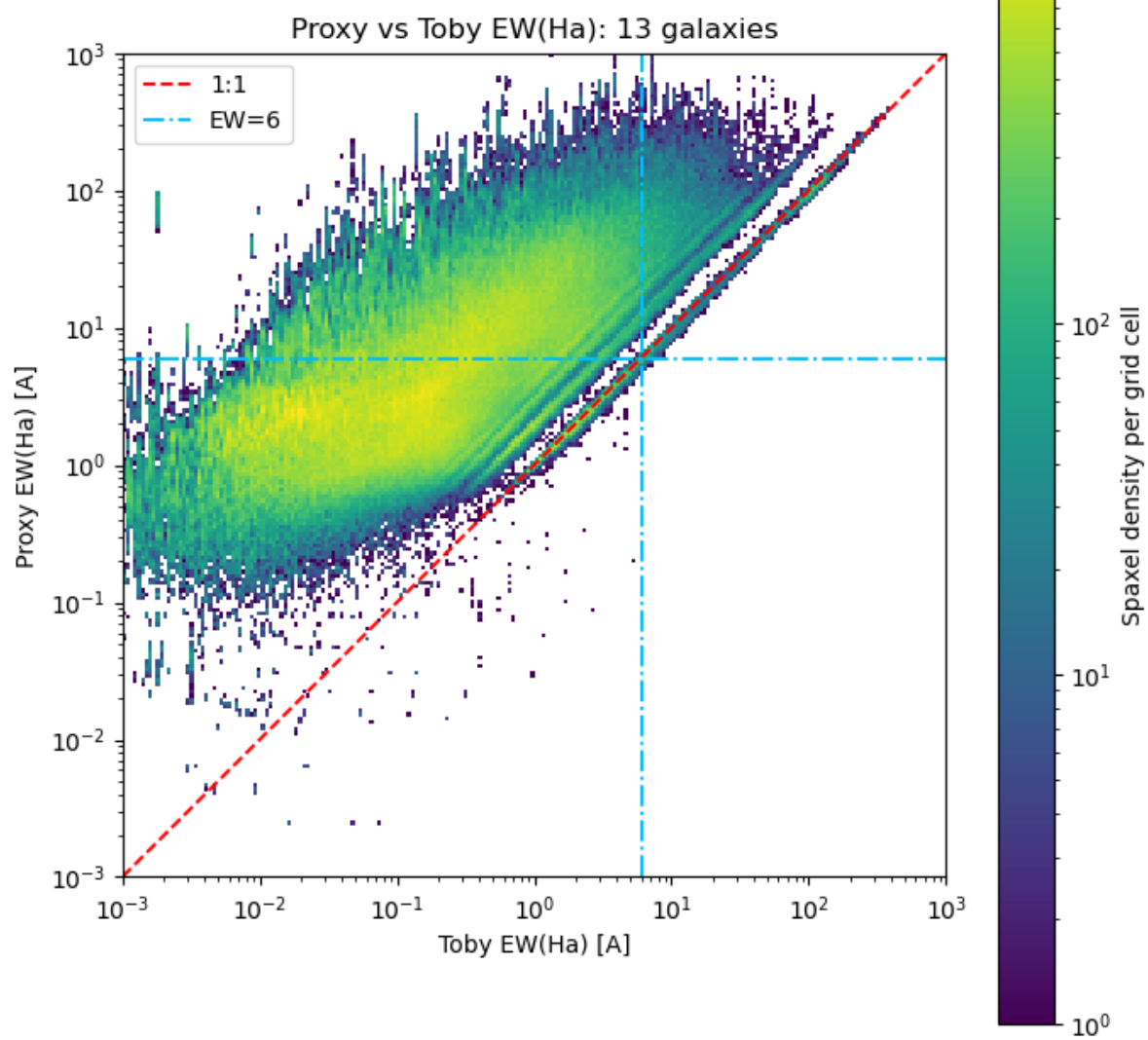
Note that $\text{EW}_{\text{proxy}}(\text{H}\alpha)$ is now based on the continuum-only spectrum measured directly across the $\text{H}\alpha$ EW window in the rest frame, whereas $\text{EW}_{\text{pseudo}}(\text{H}\alpha)$ is the older single-band approximation from the broadband R flux.

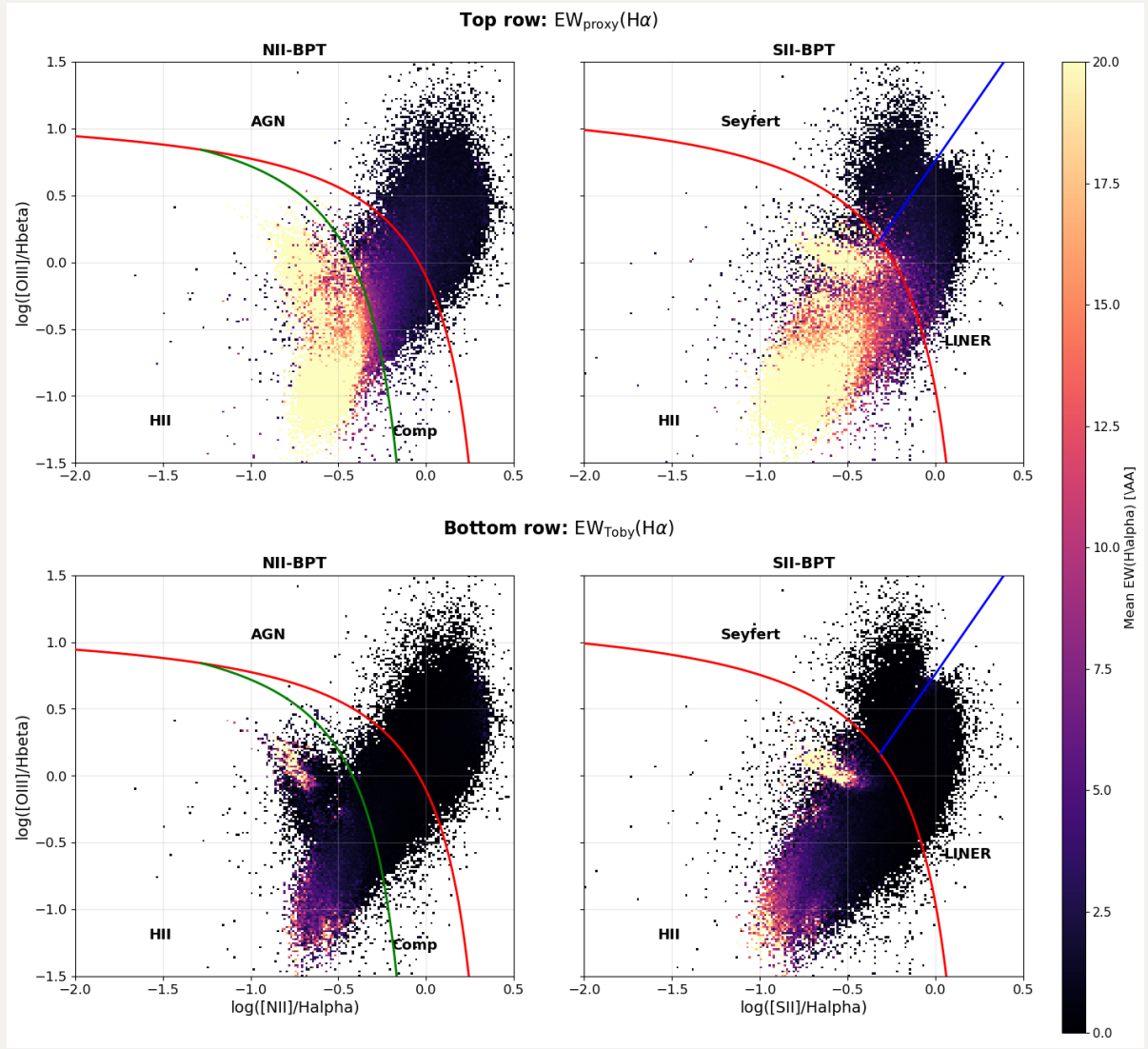
2. Comparison with $\text{EW}_{\text{pseudo}}(\text{H}\alpha)$

So the visual check looks fine. Both $\text{EW}_{\text{proxy}}(\text{H}\alpha)$ and $\text{EW}_{\text{pseudo}}(\text{H}\alpha)$ are very identical.



3. Comparison with Toby's $\text{EW}(\text{H}\alpha)$





4. Plots in paper with new $EW_{\text{proxy}}(\text{H}\alpha)$

